The AMC Data Program: Five Collectors to Start Now

Why the next round of research starts with data capture, not models — and what each stream is worth to a scrap & coin dealer

Prepared 2026-07-12. Companion to “Fed-Surprise Price Risk and AMC's Metal Inventory” (the Phase 5 business translation) and to the Phase 7 research portfolio. Canonical text: results/amc_data_acquisition_program.md.

Bottom line

The most valuable inputs to AMC's next round of research share one property: **they cannot be backfilled**. Coin premiums, search interest, futures open interest, and AMC's own transaction records exist only if someone records them as they happen. Each collector below is one to three days of build — about seven to ten working days in total — and **every week of delay is a week of irreplaceable data lost**. Three of the five deliver useful business information in their first week, before any model exists.

About this program

AMC buys scrap gold, silver, platinum, and palladium — assaying each lot for fine-metal content — and deals in gold coin and specie, so it is structurally long physical metal over a days-to-weeks holding float. A structured brainstorm (2026-07-12) produced a ten-project research portfolio for AMC: spread-floor models, hedging playbooks, crisis indices, liquidation alarms, demand forecasting. Under adversarial review, nearly every high-value project traced back to the same bottleneck: **a data series that does not exist yet and cannot be reconstructed later** — either because history is never published, is rewritten after the fact, or sits behind a paywall. This note specifies those series as five small, scheduled “collectors” feeding the project's research database.

Collector 1 — AMC's own ledger (highest priority)

What it is. A database schema and importer for AMC's transaction records, exported weekly from the point-of-sale or books: scrap lots (date, metal, gross weight, assayed fineness, fine ounces, price paid, spot at purchase, and disposition date/type/proceeds), coin trades (date, side, product, quantity, unit price, spot — hence realized premium), and, optionally, daily till counts (walk-ins, offers made, accepted).

Why now. Every research question that matters to AMC ends at this join. Without it, research can only price risk per unit of metal; with it, risk becomes dollars on AMC's actual book. The float-duration distribution — how long metal really sits — is the single number that converts the Phase 5 event findings into hedge sizes, and realized premiums are ground truth no scraped benchmark can replace.

Day-one value. Descriptive answers before any model: which metals sit longest, realized margin per lot by size and metal, premium earned by product, the float-duration histogram.

Build. One migration plus a validating importer; everything stays on the local machine — no cloud service touches AMC's books. One to two days, plus the export setup on AMC's side.

Collector 2 — Retail coin-premium panel

What it is. A once-daily capture of posted retail prices for a fixed basket of benchmark products — 1 oz American Gold Eagle and Silver Eagle, 1 oz Canadian Maple Leaf, 90% “junk” silver, generic rounds and bars — from two large online dealers (e.g., APMEX and JM Bullion): ask, buyback bid where published, spot at capture, timestamp; hence premium over melt per product per day.

Why now. No reliable free history of retail premiums exists; Internet Archive snapshots are sporadic and biased toward famous episodes, so they can validate but never train. Premium blowouts — September 2015 rationing, March 2020, February 2021, March 2023 — are exactly the episodes AMC monetizes, and the next one enters the dataset only if the collector is already running.

Day-one value. A daily benchmark: are AMC's coin prices and buyback spreads in line with the national online market? Pricing intelligence with no modelling at all.

Build. A polite scraper — a handful of pages, once daily, respecting site terms and robots.txt, internal research use — plus a breakage alert (retailers redesign pages; the collector must fail loudly). One to two days, plus occasional maintenance.

Collector 3 — Search-interest archiver (Google Trends)

What it is. A weekly (daily during spikes) pull of a fixed set of search terms — “sell gold,” “cash for gold,” “gold price,” “sell silver,” “coin shop near me” — for the United States, storing the raw response and pull timestamp.

Why now. The subtlest of the five: Google Trends **rescales its index on every request**, so a series downloaded later is not what a real-time observer saw, and a model trained on retro-pulled data quietly overstates itself. The only honest history is an archive of as-pulled snapshots — it starts accruing value the day it is switched on, and not one day before. Setup-time history is kept but permanently flagged “not captured in real time.”

Day-one value. A same-week gauge of public interest in selling — the best available leading indicator for walk-in scrap volume.

Build. A scheduled pull stored verbatim with request parameters. About one day.

Collector 4 — Futures open-interest collector (CME)

What it is. A daily capture of settlement volume and open interest — the number of futures contracts outstanding — per metals contract from the Chicago Mercantile Exchange's (CME's) public daily reports.

Why now. The project's free price source has unreliable volume for platinum and palladium (roughly 40% zero readings), and historical daily open interest is paywalled — but each day's figures are public as they appear. Forward capture is free; hindsight is not.

Day-one value. Little by itself — this one is pure seed corn. Within months it enables the “shadow positioning” nowcast (where speculative positioning stands today, versus the weekly government report that arrives three to ten days stale) and the labels for the platinum/palladium liquidation alarm — open-interest contraction is the signature of the forced-selling episodes that hit platinum-group metals (PGM) inventory hardest.

Build. Parse the CME daily bulletin or settlements pages into one table. About one day.

Collector 5 — Event calendars and Fed-surprise upkeep

What it is. Three small jobs: (a) add the 2024–26+ meeting calendar of the Federal Open Market Committee (FOMC — the Federal Reserve committee that sets interest rates) to the events table; (b) add the Bureau of Labor Statistics (BLS) release calendar for the Consumer Price Index (CPI — the main U.S. inflation report); (c) extend the monetary-surprise series — which ends December 2023, leaving the research's most actionable finding inoperable live — using the FOMC-day change in the 2-year Treasury yield, a standard stand-in already in the database; then score each new meeting the same evening.

Why now. The Phase 5 anchor finding (a hawkish Fed surprise costs gold roughly 1.4% and silver 2.9% over the following week) is only usable if each meeting can be scored as it happens. This collector is partially backfillable — but it gates the fastest, highest-value quick win in the portfolio.

Day-one value. A per-meeting hedge playbook that is live again — including for the current rate-cutting cycle the original data never covered.

Build. Calendars: hours. Surprise extension and re-estimation: about three days — the first analysis job on the new plumbing.

What unlocks what

Collector	Effort	Feeds (portfolio project)	AMC decision improved
1. AMC ledger	1–2 d + AMC export	Inventory VaR in dollars; scrap-wave target; premium ground truth	Hedge sizing, spread floors, staffing
2. Premium panel	1–2 d	Tightness-index validation; premium model; crisis playbook	Coin pricing, buyback spreads
3. Trends archiver	~1 d	Scrap-inflow surge model	Cash and assay staffing
4. CME open interest	~1 d	Shadow positioning; PGM liquidation alarm	PGM float caps, de-risking triggers
5. Calendars + Fed surprises	~0.5 d + 3 d	Live FOMC hedge playbook; event-vol risk card	Event hedging

VaR = Value-at-Risk, a standard measure of how much a position could plausibly lose over a horizon at a given confidence level.

Operations notes

- **One database.** Every collector lands in the existing research database through a numbered migration, append-only, timestamps in UTC (Coordinated Universal Time), with source and pulled-at provenance columns on every row.
- **Real-time honesty stamps.** Any row captured after the fact carries a permanent flag, and models treat flagged history as second-class — the same look-ahead discipline the research already enforces.
- **Fail loudly.** Scrapers rot; each collector alerts on schema drift, empty responses, or a missed day. A silent three-month hole discovered next year is precisely the loss this program exists to prevent.
- **Scraping etiquette.** Low frequency, a handful of pages, identified user agent, respect for robots.txt and site terms, internal research use only.
- **Backup becomes urgent.** One laptop currently holds the sole copy of the research corpus; these collectors add data that is irreplaceable by construction. An off-site backup should land in the same

sprint as the first collector.

Suggested sequence

(1) Ledger schema and the export agreement with AMC — longest business-side lead time, so start the conversation first. (2) Premium panel and Trends archiver in the same week — the clock is ticking on both. (3) Calendars — an hour, unblocks the event work. (4) CME open interest. (5) The Fed-surprise extension as the first analysis job — the fastest route to a live, business-usable deliverable. Roughly seven to ten working days in total, mostly parallelizable.

What these collectors do not do

- **They are not models.** Value compounds with time; most downstream models need six to twenty-four months of accrued labels before honest evaluation is possible. The point of starting now is that the clock only runs forward.
- **Posted online premiums are not AMC's premiums.** The panel is a national benchmark; AMC's local market, mix, and condition grading differ. The ledger is the ground truth.
- **Search interest is relative, not absolute.** It indicates surges, not volumes.
- **No collector predicts anything.** Each records reality cheaply so later models can be tested honestly — the same evaluation-first discipline behind the Phase 5 findings.